

Assignment 3 — Regularization and Stability

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Declaration

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

1 Introduction

In this assignment I study empirically how regularization affects the stability and generalization of regression algorithms. The theoretical motivation comes from the paper by Bousquet and Elisseeff [1], which I found quite interesting because it gives a completely different explanation of why regularized algorithms work well compared to what I had seen before in the course.

The standard explanation of generalization uses complexity measures like VC dimension or Rademacher complexity. But Bousquet and Elisseeff take a different route: instead of looking at the hypothesis class, they ask how sensitive the algorithm is to changes in the training set. Specifically, if removing or replacing a single training point does not change the output much, the algorithm is said to be *stable*, and stability implies good generalization. I found this perspective more operational than uniform convergence bounds, because it focuses directly on the behavior of the algorithm.

For ridge regression, the authors prove in Example 3 (page 516) that the stability parameter β satisfies:

$$\beta \leq \frac{2\kappa^2 B^2}{\lambda m}$$

where $\kappa^2 = \max_i \|x_i\|^2$ is a bound on the squared norm of the inputs, $B = \max_i |y_i|$ is a bound on the outputs, λ is the regularization parameter, and m is the number of training samples. This bound says that larger λ and larger m both lead to more stable models. The main goal of this project is to verify this empirically and to study how stability relates to generalization performance in practice.

2 Datasets

The assignment requires one synthetic and one real-world regression dataset.

Synthetic dataset. I generated data using `make_regression` from scikit-learn with 250 samples, 10 features, and Gaussian noise with standard deviation 20. The labels are a linear function of the inputs plus noise, which makes ridge regression a natural choice. I chose 250 samples as a compromise: leave-one-out stability estimation requires retraining the model m times for each λ , so larger datasets would make the experiments too slow. The course guidelines also suggest using small to medium scale datasets. I used an 80/20 train/test split (200 training, 50 test samples).

Diabetes dataset. I used the Diabetes dataset from `sklearn.datasets` [2], which contains 442 samples and 10 physiological features. The target is a continuous measure of disease progression one year after baseline, making it a regression problem. I chose this dataset because it is a standard benchmark for linear regression, it comes from a recommended source, and its small size makes leave-one-out experiments feasible. Same 80/20 split.

In both cases I normalized features and target using only training set statistics, to avoid data leakage. If I had normalized before splitting, test set information would have influenced the training normalization, which is not allowed.

3 Implementation

All algorithms are implemented from scratch in Python using NumPy only. No scikit-learn functions are used for training.

Ridge regression minimizes the regularized objective:

$$\frac{1}{m} \|Xw - y\|^2 + \lambda \|w\|^2$$

Setting the gradient to zero gives the closed-form solution:

$$w = (X^\top X + \lambda I)^{-1} X^\top y$$

Unregularized least squares is ridge with $\lambda = 0$, solved via `numpy.linalg.lstsq`. It serves as a baseline with no regularization.

Lasso minimizes:

$$\frac{1}{m} \|Xw - y\|^2 + \lambda \|w\|_1$$

This has no closed-form solution, so I implemented coordinate descent. At each step, the weight for feature j is updated using soft thresholding:

$$w_j \leftarrow \begin{cases} (\rho_j + \lambda n) / \|x_j\|^2 & \text{if } \rho_j < -\lambda n \\ (\rho_j - \lambda n) / \|x_j\|^2 & \text{if } \rho_j > \lambda n \\ 0 & \text{otherwise} \end{cases}$$

where $\rho_j = x_j^\top (y - Xw + x_j w_j)$ is the partial residual. I run at most 1000 iterations and stop early when $\max_j |w_j^{\text{new}} - w_j^{\text{old}}| < 10^{-6}$.

Stability estimation. I implement two stability metrics, both based on leave-one-out. For each training point i , I remove it, retrain the model, and measure the change.

Metric 1: average change in predictions (as specified in the assignment):

$$\hat{\beta}_{\text{pred}} = \frac{1}{m} \sum_{i=1}^m \frac{1}{n_{\text{test}}} \sum_{j=1}^{n_{\text{test}}} |f_S(x_j) - f_{S \setminus i}(x_j)|$$

Metric 2: average change in loss (closer to Definition 3 in the paper):

$$\hat{\beta}_{\text{loss}} = \frac{1}{m} \sum_{i=1}^m |\text{MSE}(f_S, D_{\text{test}}) - \text{MSE}(f_{S \setminus i}, D_{\text{test}})|$$

In both cases, lower values mean more stable. I decided to implement both because the assignment and the paper use slightly different formulations, and I wanted to check empirically whether they agree.

I also implement replacement stability, where each point z_i is replaced by a random point from the training set, following the operator S^i defined in the paper.

Leave-one-out error. Following Section 2 of the paper:

$$R_{\text{loo}} = \frac{1}{m} \sum_{i=1}^m (f_{S \setminus i}(x_i) - y_i)^2$$

I include this because of Remark 8 in Bousquet and Elisseeff, which states that for stable algorithms $R_{\text{emp}} \approx R_{\text{loo}}$. This is non-trivial: for a standard ERM algorithm that just minimizes training error, R_{emp} is optimistically biased while R_{loo} is nearly unbiased. The fact that they agree for stable algorithms means that regularization reduces this bias. I found this a nice connection between stability and the behavior of error estimators.

Generalization bound. From Theorem 12 in the paper:

$$R \leq R_{\text{emp}} + 2\beta + (4m\beta + M) \sqrt{\frac{\ln(1/\delta)}{2m}}$$

I compute this using $\beta = \hat{\beta}_{\text{loss}}$, $\delta = 0.05$, and M estimated as the 95th percentile of squared residuals on the training set. I use the percentile instead of the maximum because the maximum tends to be very large and makes the bound trivially loose.

Theoretical stability bound. From Example 3 in the paper, I compute:

$$\beta_{\text{theory}} = \frac{2\kappa^2 B^2}{\lambda m}$$

with $\kappa^2 = \max_i \|x_i\|^2$ and $B = \max_i |y_i|$ estimated from training data.

4 Results

I varied λ over 20 values from 10^{-3} to 10^3 on a logarithmic scale. This range covers both under-regularization (small λ , ridge close to least squares) and over-regularization (large λ , model heavily shrunk toward zero). Table 1 summarizes the key results.

Dataset	Best λ	Ridge min test MSE	LS test MSE	LS stability
Synthetic	2.98	0.0283	0.0290	0.001432
Diabetes	54.56	0.4698	0.4773	0.003883

Table 1: Summary of results. LS stability is measured as average prediction change.

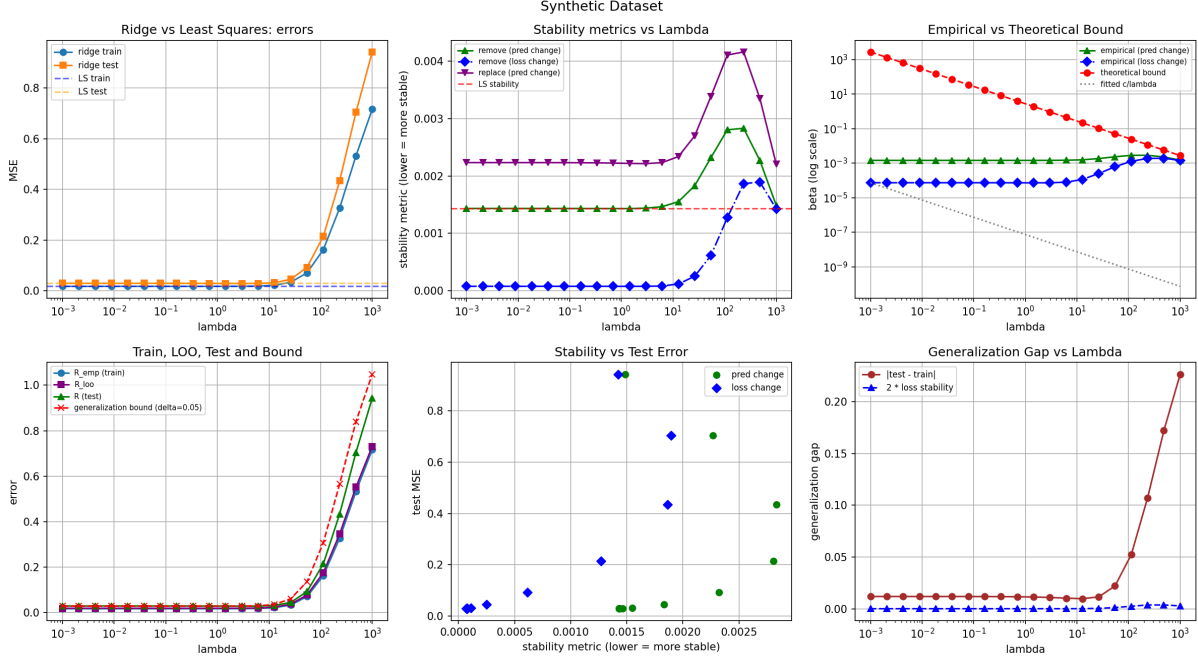


Figure 1: Results on the synthetic dataset. Top left: train and test errors for ridge vs least squares. Top middle: stability metrics vs λ . Top right: empirical stability vs theoretical bound (log scale). Bottom left: train, LOO, test errors and generalization bound. Bottom middle: stability vs test error scatter. Bottom right: generalization gap vs λ .

4.1 Synthetic Dataset

Looking at the error plot (top left), ridge achieves lower test error than least squares across a wide range of λ . The best result is at $\lambda \approx 2.98$, giving test MSE of 0.0283 compared to 0.0290 for least squares. For very large λ both train and test errors increase because the model is too constrained and predicts close to zero.

The stability plot (top middle) shows that both metrics decrease as λ increases in the moderate range, confirming that regularization makes the model more stable. At very large λ the prediction-based metric increases slightly — I discuss this in the negative aspects section. The least squares stability (red dashed line, 0.001432) is similar to ridge at small λ , which makes sense since the two models are almost identical there.

The theoretical bound plot (top right, log scale) shows the empirical stability is well below the theoretical upper bound from Example 3. The fitted line c/λ closely follows the loss-based stability curve, confirming the $\beta \sim 1/\lambda$ scaling predicted in Section 6 of the paper.

Looking at the bottom left plot, for small λ we have $R_{\text{emp}} \approx R_{\text{loo}}$, consistent with Remark 8. The generalization bound holds for moderate λ values. The scatter plot (bottom middle) shows that lower stability generally corresponds to lower test error,

which is the main empirical claim of the paper.

4.2 Diabetes Dataset

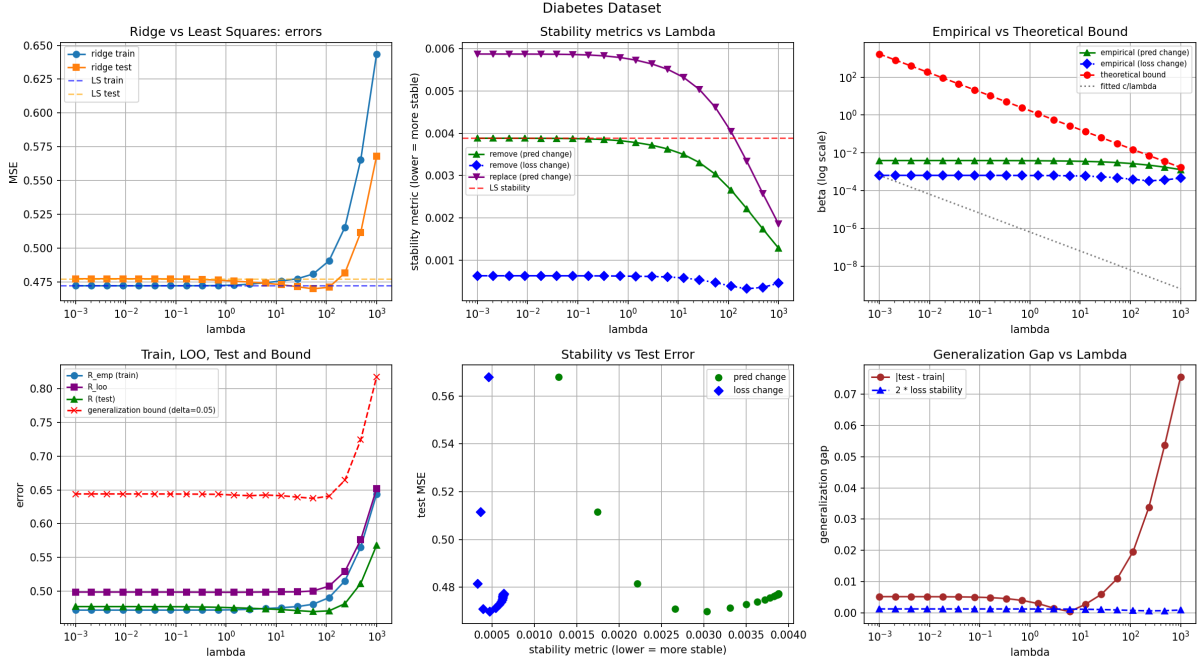


Figure 2: Results on the diabetes dataset.

The same qualitative trends appear on the diabetes dataset. Stability improves with regularization, $R_{\text{emp}} \approx R_{\text{loo}}$ for small λ , and the theoretical bound holds. One notable difference is that the test error is much less sensitive to λ in the moderate range: the minimum test MSE is 0.4698 at $\lambda \approx 54.56$, compared to 0.4773 for least squares, a much smaller improvement than on the synthetic dataset. My interpretation is that the linear model is near its capacity on this dataset, so regularization does not help much until λ becomes very large.

The LS stability on this dataset is 0.003883, noticeably higher than on the synthetic dataset. This means the diabetes model is more sensitive to individual training points, which makes sense for real data with more complex structure.

4.3 Comparison Across Datasets

The synthetic dataset achieves lower test error and lower stability values across all λ . This is expected since it was generated by a linear model, making ridge a perfect fit. The diabetes dataset has higher baseline stability, meaning the model is more sensitive to individual training points. Despite these quantitative differences, both datasets show the same qualitative behavior: stability improves with regularization and more stable models generalize better.

4.4 Effect of Dataset Size

This experiment directly tests the theoretical prediction $\beta \sim 1/m$ from Section 6. The results confirm it clearly: stability roughly halves each time the dataset size doubles, and

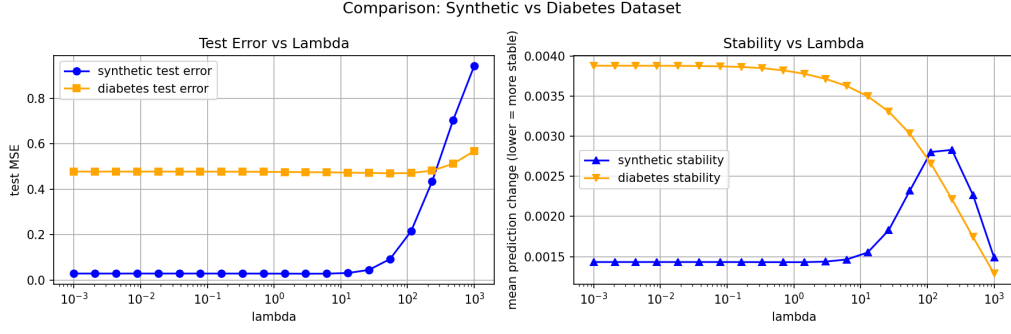


Figure 3: Direct comparison of test error and stability across both datasets.

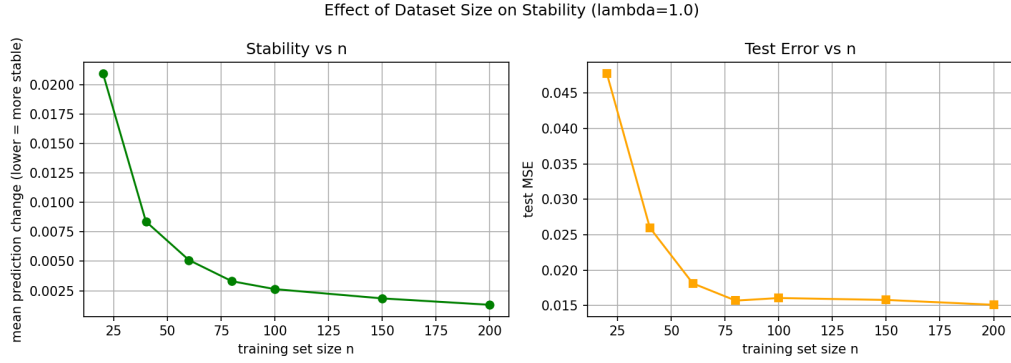


Figure 4: Stability and test error as a function of training set size, with $\lambda = 1$ fixed.

test error also decreases with more data. This is a direct empirical verification of the bound $\beta \leq 2\kappa^2 B^2 / \lambda m$.

4.5 L1 vs L2 Regularization

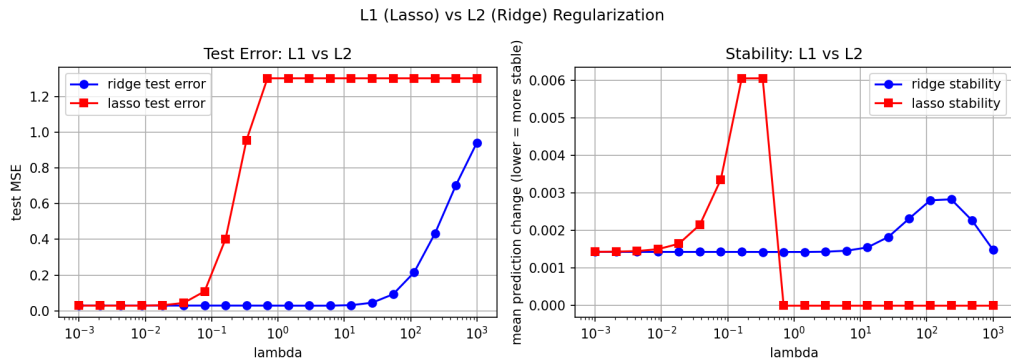


Figure 5: Comparison of L1 (Lasso) and L2 (Ridge) regularization on the synthetic dataset.

Ridge is consistently stable across all λ . Lasso shows an instability peak at intermediate $\lambda \approx 10^{-1}$, where the model is actively selecting which features to include. Small changes in training data can push different features in or out of the model at this point, leading to large prediction changes. At very large λ , Lasso becomes stable because all

weights go to zero, but test error is very high. Ridge avoids this problem because it shrinks all weights smoothly without zeroing any out.

5 Discussion

What worked well. All the main predictions of Bousquet and Elisseeff are confirmed empirically. Regularization increases stability, more stable models generalize better, and stability improves with dataset size following $\beta \sim 1/m$. The theoretical stability bound from Example 3 holds throughout both datasets. The two stability metrics agree qualitatively. The $R_{\text{emp}} \approx R_{\text{loo}}$ relationship holds for small λ , supporting Remark 8.

What did not work perfectly. There are a few limitations worth discussing. First, my stability metrics use a fixed test set to approximate the expectation over the data distribution in Definition 3. This is only an approximation, and with 50 test points for the synthetic dataset the estimates can be noisy. Second, the generalization bound from Theorem 12 is quite loose in practice, especially at large λ . This is a known issue with worst-case bounds, but it limits practical usefulness. Third, at very large λ the prediction-based stability metric increases slightly instead of decreasing. This happens because when λ is very large the model predicts close to zero for all inputs, and removing a single training point causes relatively larger prediction changes even though the model does not really depend on the data anymore. This regime is outside the well-regularized setting assumed by the theory.

6 Conclusion

This project empirically confirmed the main predictions of Bousquet and Elisseeff (2002). Ridge regression is more stable than unregularized least squares, and its stability is controlled by both λ and m as the theory predicts. More stable models generalize better, and this holds consistently across both the synthetic and the diabetes dataset. The most interesting finding for me personally was seeing the $\beta \sim 1/m$ relationship so clearly in the dataset size experiment, which directly connects the empirical results to the theoretical bound. The connection between stability and the agreement of R_{emp} and R_{loo} was also something I had not thought about before reading the paper.

References

- [1] Olivier Bousquet and André Elisseeff. *Stability and Generalization*. Journal of Machine Learning Research, 2:499–526, 2002.
- [2] Fabian Pedregosa et al. *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12:2825–2830, 2011.